Curriculum Vitae

# Alon Gurman

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**CONTACT**

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**EMPLOYMENT**

Postdoctoral Researcher (Joint Rutgers-Flatiron Postdoctoral Fellow)
Department of Physics and Astronomy, Rutgers University, New Jersey, 2026-present

**EDUCATION**

*Ph.D. in Physics and Astronomy*
Tel Aviv University, Tel Aviv, Israel, 2020-2026

*B.Sc. in Mathematics and Physics*
Tel Aviv University, Tel Aviv, Israel, Magna Cum Laude, 2016-2020

*Visiting Ph.D. Student*
Center for Computational Astrophysics, Flatiron Institute, 2024-present

**PUBLICATIONS**

**Gurman, A.**, Hu, C. Y., Grudic, M. Y., Steinwandel, U. P., Sternberg, A. "The GHOSDT Simulations: II. Missing Molecules in Simulations of a Self-Regulated Interstellar Medium". *ApJ, 995, 186 (2025).*

**Gurman, A.**, Sternberg, A., Bialy, A., Cochrane, R. K., Stern, J. "Molecular Hydrogen in High-redshift Damped Lyman-$\alpha$ Absorbers". *ApJ, 995, 116 (2025).*

**Gurman, A.**, Steinwandel, U. P., Hu, C. Y., Sternberg, A. "The GHOSDT Simulations: I. Magnetic Support in Gas Rich Disks". *ApJ, 984, 142 (2025).*

**Gurman, A.**, Hu, C. Y., Sternberg, A., van Dishoeck, E. F., "[CII] Emission in a Self-Regulated Interstellar Medium". *ApJ, 965, 179 (2024).*

Sternberg, A., Bialy, S., **Gurman, A.**, "HI in Molecular Clouds: Irradiation by FUV Plus Cosmic Rays". *ApJ, 960, 8 (2024).*

Sternberg, A., **Gurman, A.**, Bialy, S., "HI-to-H2 Transitions in Dust-free Interstellar Gas". *ApJ, 920, 83 (2021).*

**ACADEMIC HONORS**

Tel Aviv University scholarship for excellence in teaching, 2024

Tel Aviv University scholarship for excellence in teaching and research, 2022

Tel Aviv University scholarship for outstanding academic performance in graduate studies, 2021

**SEMINAR TALKS**

Journal Club Talk, University of California San Diego, San Diego, CA, 2025
Institute for Theory and Computation Luncheon, Harvard-Smithsonian Center for Astrophysics, Cambridge, MA, 2025

NY Area Galaxy Seminar, Center for Computational Astrophysics, New York, 2025
University of Illinois Urbana-Champaign, Illinois, 2025
CCA Coffee Talks, Flatiron Institute, New York, 2025
Stars and Interstellar Medium group seminar, Center for Computational Astrophysics, Flatiron Institute, New York, NY, 2024
Star Formation/ISM Rendezvous seminar, Princeton University, Princeton, NJ, 2024
Astrophysics Seminar, Tel Aviv University, Tel Aviv, Israel, 2024
Astrophysics Seminar, Technion – Israel Institute of Technology, Haifa, Israel, 2024
Galaxy Formation Seminar, Center for Computational Astrophysics, New York, 2023

**CONFERENCE TALKS**

GIZMOfest2026 Workshop, Center for Computational Astrophysics, Flatiron Institute, New York, 2026
Geometry of Flows Meeting, New York University, New York, 2026
The Magnetized Turbulent Universe, Playa del Carmen, Mexico, 2025
Far-Infrared Fine-Structure Line Workshop, Winona, Minnesota, 2025
VICO-CICO-CCA Cosmic Origins Workshop, Flatiron Institute, New York, 2024
ASCOS - Astronomy and Cosmology Conference, Weizmann Institute of Science, Rehovot, Israel, 2024

**TEACHING AND MENTORING**

Teaching Assistant, Introduction to Astrophysics (physics final year), 2020-2025
Teaching Assistant, 2nd year physics lab, 2020-2025
Co-advisor, Matan Berkovich, Technion – Israel Institute of Technology, Haifa, Israel, 2025-present
Professional Development for Teachers and Development of Teaching Materials, joint program by Tel Aviv University, Israeli Space Agency, and the Israeli Ministry of Education, 2021-2023
Mentor, Meduyakot program for women in exact sciences, Tel Aviv Unviersty, 2021

**PROFESSIONAL EXPERIENCE**

Reviewer in Training, Monthly Notices of the Royal Astronomical Society, 2023-present
Research Assistant, MRI motion correction with Frydman Group, Weizmann Institute of Science, 2018-2019

**CLIMBING AND MOUNTAINEERING**

Traveled across the Middle East, Europe, and USA, climbing big walls at elite level such as El Capitan in Yosemite National Park CA, Mt. Hooker in the Wind River Range WY, and Djebel Nasrani in Wadi Rum, Jordan
U19 sport climbing Israeli national champion, 2011